JSP

Sr. No.	Area	Topic	What to Cover	Technologies / Practices
1	Web Foundations	Role of JSP	Where JSP fits in Java web stack, comparison with Servlets	Java EE / Jakarta EE
2	Web Foundations	Request–Response Lifecycle	How a request reaches JSP, JSP-to-Servlet translation	HTTP, Servlet Container
3	JSP Basics	JSP Syntax	Scriptlets, expressions, declarations (and why they are discouraged)	JSP
4	JSP Basics	Implicit Objects	request, response, session, application, out	JSP
5	JSP Basics	Directives	page, include, taglib directives and their effects	JSP
6	JSP Basics	JSP Compilation Model	How JSP is compiled into a Servlet, lifecycle methods	Servlet API
7	MVC Concepts	JSP in MVC	JSP as View layer only, avoiding business logic	MVC Pattern
8	JSTL Core	Purpose of JSTL	Why JSTL exists, replacing scriptlets	JSTL
9	JSTL Core	Core Tags	c:if, c:forEach, c:choose, scope handling	JSTL Core
11	Expression Language	EL Basics	${}, property access, null handling	JSP EL
12	Expression Language	EL with Collections	Accessing lists, maps, and nested objects	JSP EL
14	Integration	JSP with Servlets	Forward vs redirect, request attributes	Servlet + JSP
15	Best Practices	JSP Anti-Patterns	Business logic in JSP, scriptlet-heavy pages	Clean Code

APIs

Sr. No.	API Area	Topic	What to Cover	Typical Technologies / Practices
1	Foundations	What an API Is	Contract between systems, boundaries	REST, HTTP
2	Foundations	API Types	Public, private, partner APIs	REST, internal APIs
3	Foundations	Synchronous vs Asynchronous APIs	Blocking vs event-driven	REST, webhooks
4	Foundations	API as a Product	Stability, consumers, backward compatibility	Versioning policies
5	Protocol Basics	HTTP Fundamentals	Methods, headers, status codes	HTTP/1.1, HTTP/2
6	Protocol Basics	Idempotency	Safe retries	PUT, DELETE, idempotency keys
7	Protocol Basics	Content Negotiation	Accept / Content-Type	JSON, XML
9	REST Design	Resource Modeling	Nouns vs verbs	URI design
10	REST Design	HTTP Methods Usage	GET, POST, PUT, PATCH, DELETE	REST conventions
11	REST Design	Status Codes	Correct success & error codes	2xx–5xx
13	REST Design	Consistency	Naming, casing, patterns	Style guides
14	Request/Response	Payload Design	JSON structure, envelopes	DTOs
15	Request/Response	Pagination	Offset vs cursor	Query params
16	Request/Response	Filtering & Sorting	Query parameters	URL encoding
18	Request/Response	Large Payload Handling	Chunking, limits	Streaming, pagination
19	Versioning	API Versioning Strategies	URI, header, media type	/v1, headers
22	Error Handling	Error Model	Structured error responses	errorCode, message

23	Error Handling	Client vs Server Errors	Responsibility split	4xx vs 5xx
24	Error Handling	Validation Errors	Field-level feedback	Constraint violations
25	Error Handling	Retry Semantics	When clients should retry	429, 503
26	Security	Authentication	Who are you	API keys, OAuth2
27	Security	Authorization	What can you do	Scopes, roles
28	Security	Token-Based Security	JWT structure & expiry	JWT
30	Security	Input Validation	Injection prevention	Validation layers
38	Documentation	API Documentation	Human-readable contracts	OpenAPI
39	Documentation	OpenAPI Spec	Paths, schemas, responses	Swagger
42	Client Integration	API Consumption	Calling external APIs	HTTP clients
43	Client Integration	Timeout Handling	Avoid hanging calls	Timeouts
44	Client Integration	Retry & Backoff	Resilience patterns	Exponential backoff
46	Async APIs	Webhooks	Event callbacks	HTTP callbacks
47	Async APIs	Event-Driven APIs	Publish/subscribe	Kafka, SNS
48	Async APIs	Message Contracts	Schema evolution	JSON schema
57	Observability	Logging	Request/response logs	Structured logs